

Estimation

From a Point Estimate to an Interval

You Only Ever See One Sample

You compute $\hat{\theta}$ from the one dataset you have. A colleague with a different sample from the same population gets a different number. Neither of you is wrong.

The object this whole deck is about

The **sampling distribution** of $\hat{\theta}$: the distribution of the numbers you would get by repeating the study. Every idea that follows is a statement about one of its features.

- **Judge an estimator:** bias, variance, consistency
- **Build one:** losses, maximum likelihood, GMM — and the bridges between them
- **Report it:** efficiency, confidence intervals, p-values

A point estimate is one draw from that distribution. An interval reports its width.

Where the Distribution Sits

What Makes an Estimator “Unbiased”?

Intuition: imagine repeating your study many times, each time computing $\hat{\theta}$ from a fresh sample.

- If the *average* of all those estimates equals the true θ , the estimator is **unbiased**.
- If it systematically overshoots or undershoots, it's **biased** — like a bathroom scale that always reads 2 kg too heavy.

i Formal Definition

$\hat{\theta}$ is **unbiased** for θ if

$$\mathbb{E}[\hat{\theta}] = \theta \quad \text{for all } \theta \in \Theta$$

Its **bias** is $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$.

Unbiased or Biased? Some Familiar Examples



Unbiased

- $\bar{X}$ for the mean μ
- $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ for σ^2



Biased

- $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$
- Most MLEs (coming up), in finite samples

$E[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2$ — the bias vanishes as n grows, which is the first hint that *unbiased* and *consistent* are different promises.

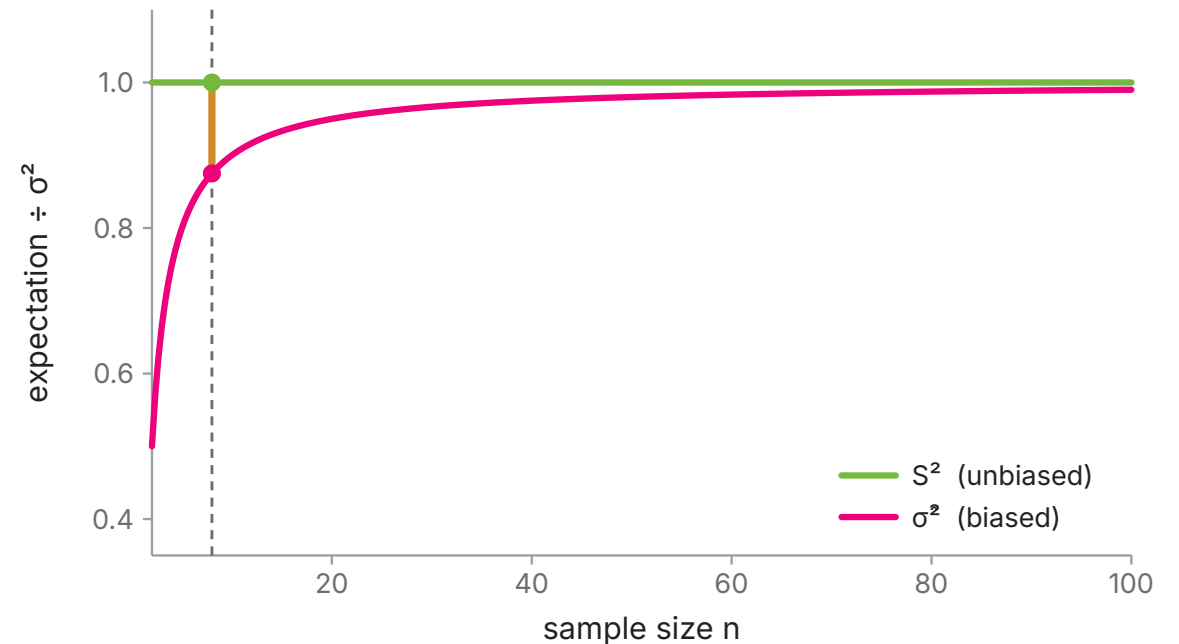
n

8



dividing by n instead of $n - 1$ shrinks the estimate

$n = 8$: bias = $-\sigma^2/n = -0.125 \cdot \sigma^2$



Unbiasedness, Visualized

repeated samples

150



sample size n

4

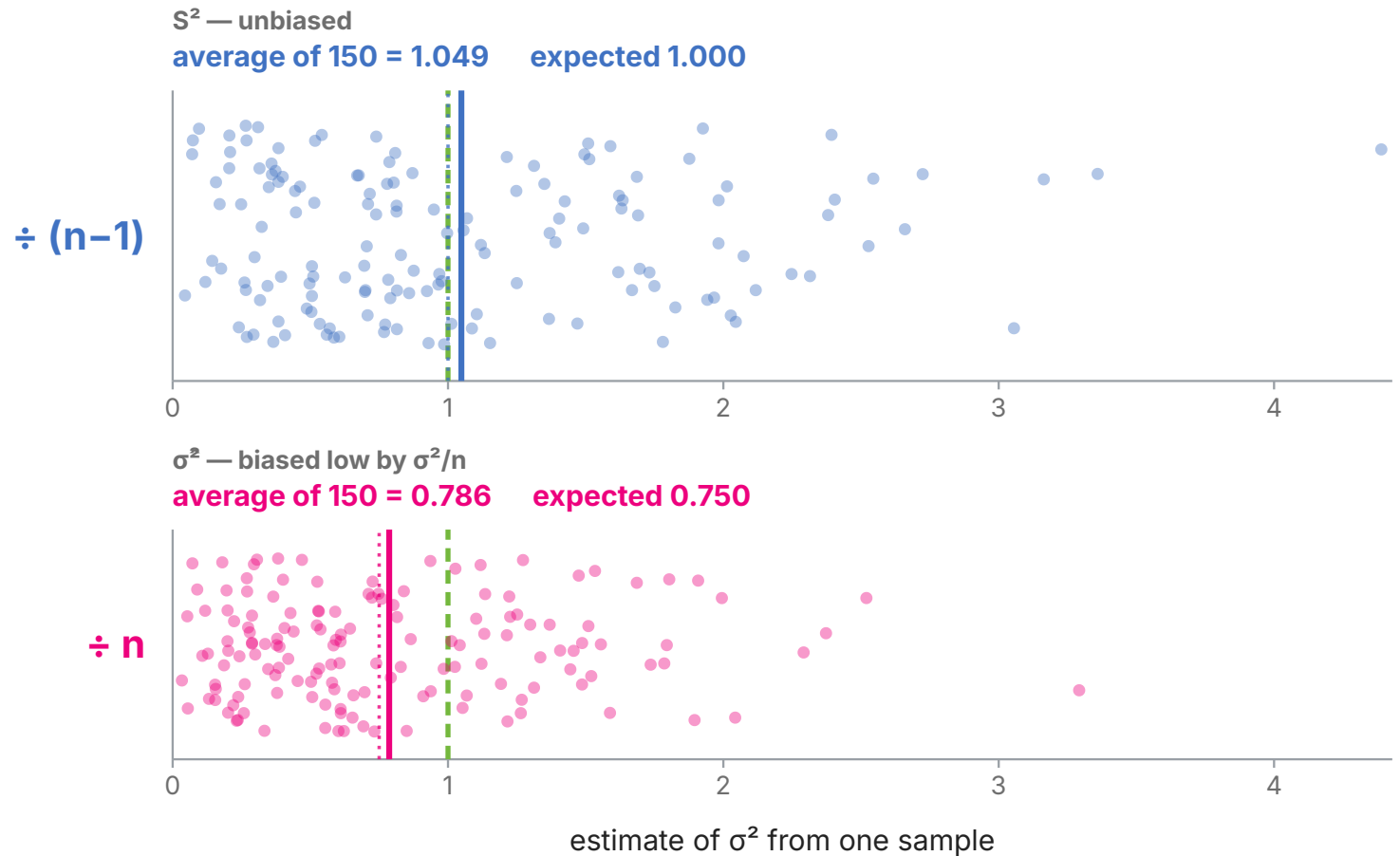


New samples

Every sample is drawn honestly from the true distribution — nothing is shifted. Both rows are handed the **same** data and differ only in the divisor.

The dotted rule is where each estimator is *expected* to land. Add repetitions and each solid average walks onto its own dotted rule — one of which is not σ^2 .

Pull the repetitions down and the averages become unreliable: unbiasedness is a statement about the long run, not about one study.



Bias vs. Variance: Two Ways to Be Wrong

Think of an estimator as arrows thrown at a target, where the bullseye is the true θ :

- **High bias:** the arrows cluster tightly, but off to one side — consistently wrong in the same way
- **High variance:** the arrows scatter widely around the bullseye — right *on average*, but unpredictable in any one sample
- An ideal estimator has *both* low bias and low variance — but in practice we often have to trade one off against the other

Bias and Variance, Visualized

shots per target

10

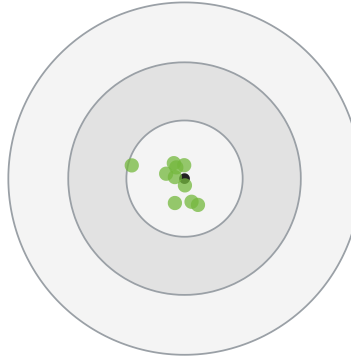


Rows differ in **bias** — whether the cloud is centred on the bullseye. Columns differ in **variance** — how tightly it is packed.

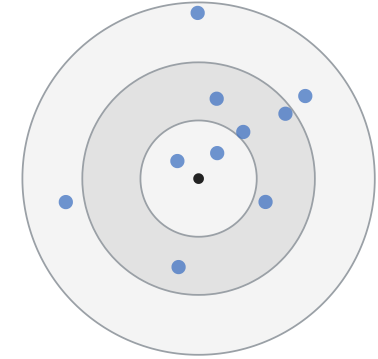
At one or two shots the four targets are indistinguishable. The pattern is a property of the *distribution*, not of any single estimate you will ever compute.

Keep in mind: each dot is an estimate, not a data point!

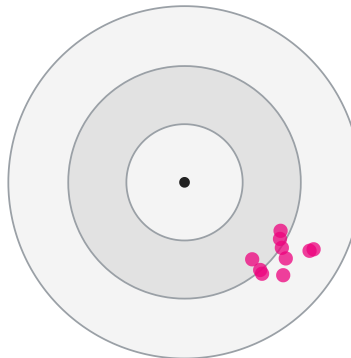
low bias, low variance



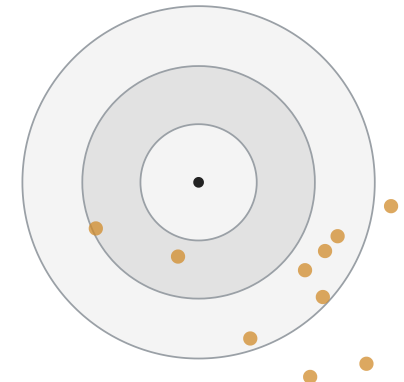
low bias, high variance



high bias, low variance



high bias, high variance



Mean Squared Error: Combining Both



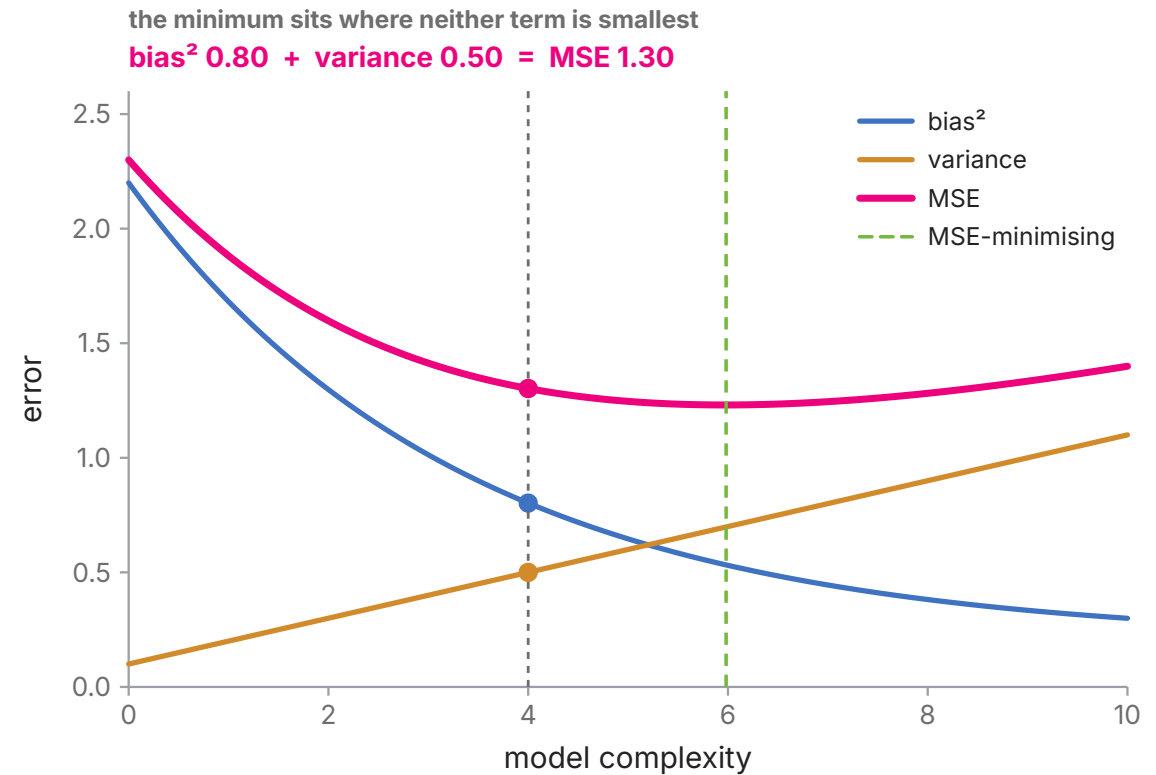
Definition and Decomposition

$$\text{MSE}(\hat{\theta}) = \underbrace{\text{Var}(\hat{\theta})}_{\text{Variance}} + \underbrace{[\text{Bias}(\hat{\theta})]^2}_{\text{Bias}^2}$$

We don't only look for unbiased estimators: a slightly biased estimator with much lower variance can have **lower MSE overall**. Drag the slider to read the trade-off at each point.

model complexity

4



Example: A Little Bias Can Pay Off

For $X_1, \dots, X_n \sim N(\mu, 1)$, compare two estimators of μ :

- $\hat{\mu}_1 = \bar{X}$ (unbiased)
- $\hat{\mu}_2 = c\bar{X}$ for some $0 < c < 1$ (biased — “shrunk” toward 0)

Their mean squared errors are:

$$\text{MSE}(\hat{\mu}_1) = \frac{1}{n}$$

$$\text{MSE}(\hat{\mu}_2) = \frac{c^2}{n} + (1 - c)^2 \mu^2$$

i Takeaway

When μ is close to 0, the biased $\hat{\mu}_2$ can have **smaller** MSE than the unbiased $\bar{X}$. This is the idea behind shrinkage estimators like Ridge regression.

A Little Bias Can Pay Off, Visualized

n

10



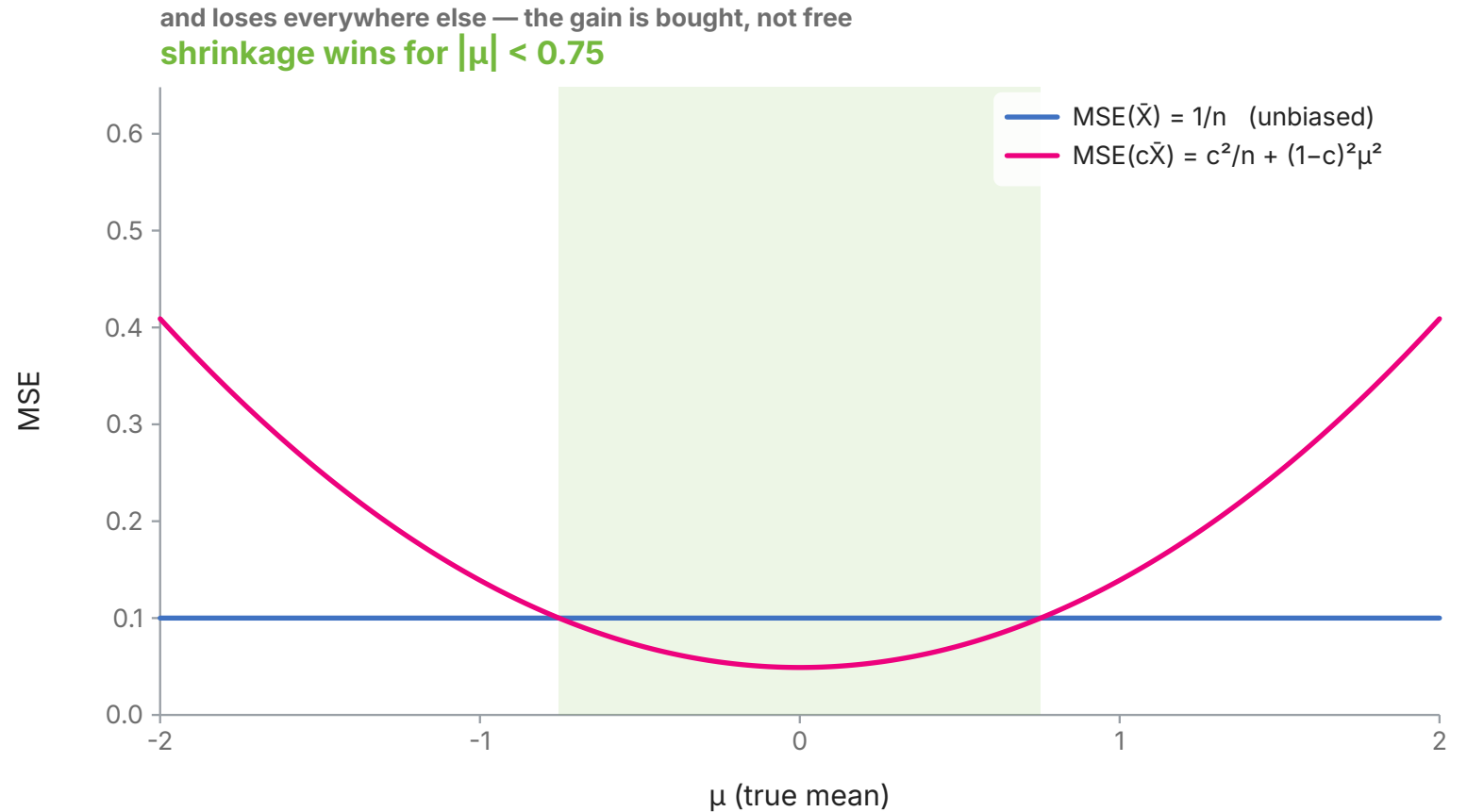
c

0.7



Shaded: where the shrunk estimator has the **lower** MSE.

Shrinking hard (c small) wins big near $\mu = 0$ and loses badly far from it. Raising n narrows the shaded band — with enough data there is little left to gain by biasing the estimate.



What More Data Buys

The Law of Large Numbers, One Path at a Time

draws n

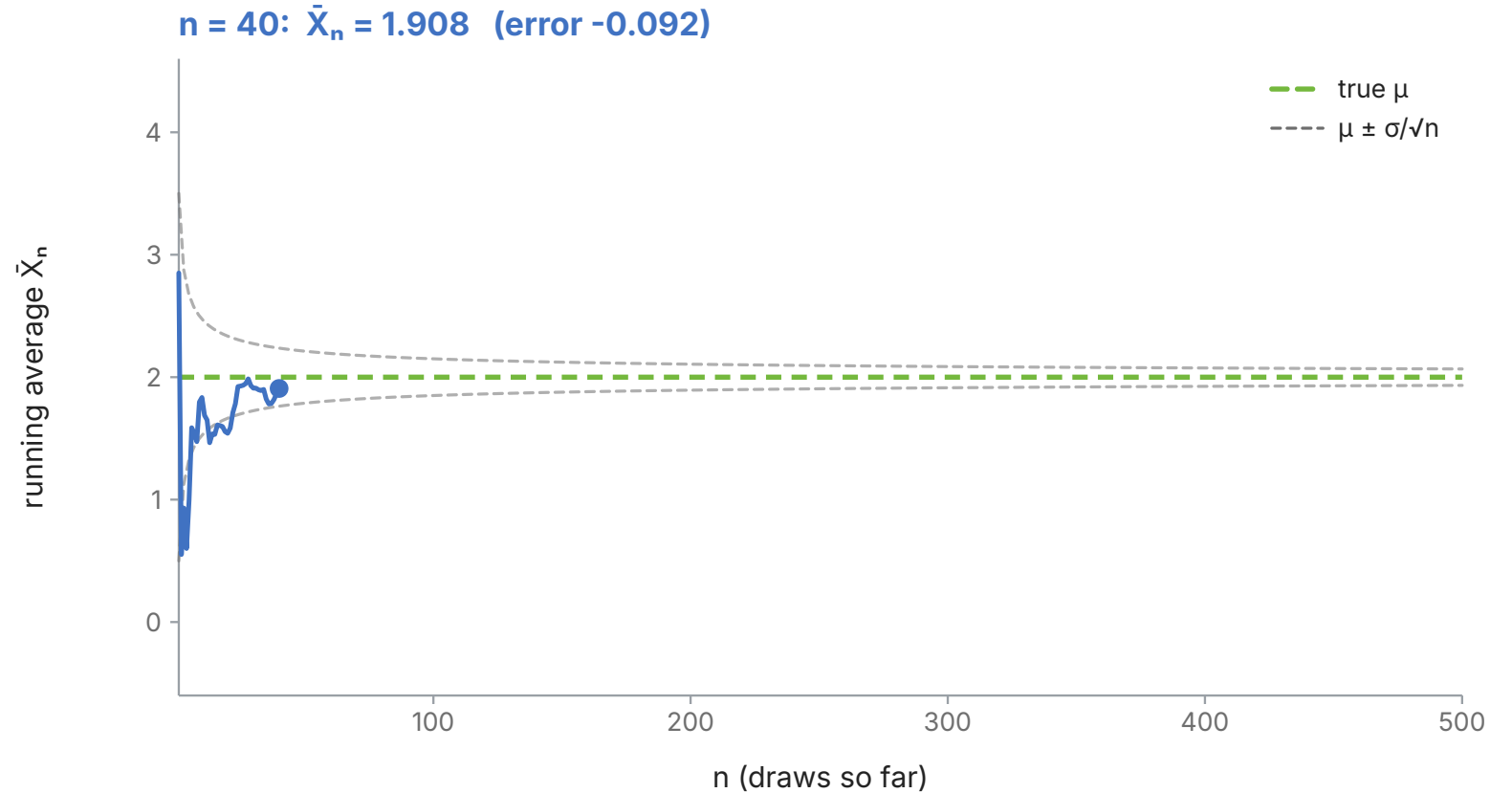
40



New path

Every property so far was stated for a *fixed* n . Now let n grow: the running average settles down near the true μ .

The dashed funnel is $\mu \pm \sigma/\sqrt{n}$ — the path is squeezed by arithmetic, not by luck. Hold on to that shape: by the end of this deck it will be an interval, drawn the other way round.



Consistency: Getting It Right *Eventually*

Unbiasedness is about being right *on average* for a given n . **Consistency** instead asks: does the estimator get closer and closer to the truth as we collect more data?

i Definition

$\hat{\theta}_n$ is **consistent** if $\hat{\theta}_n \xrightarrow{p} \theta$ as $n \rightarrow \infty$, i.e. for every $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$$

i An Easy Way to Check It

If both of these hold, $\hat{\theta}_n$ is automatically consistent:

$$\lim_{n \rightarrow \infty} \text{Bias}(\hat{\theta}_n) = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \text{Var}(\hat{\theta}_n) = 0$$

Consistency in Pictures

n

10



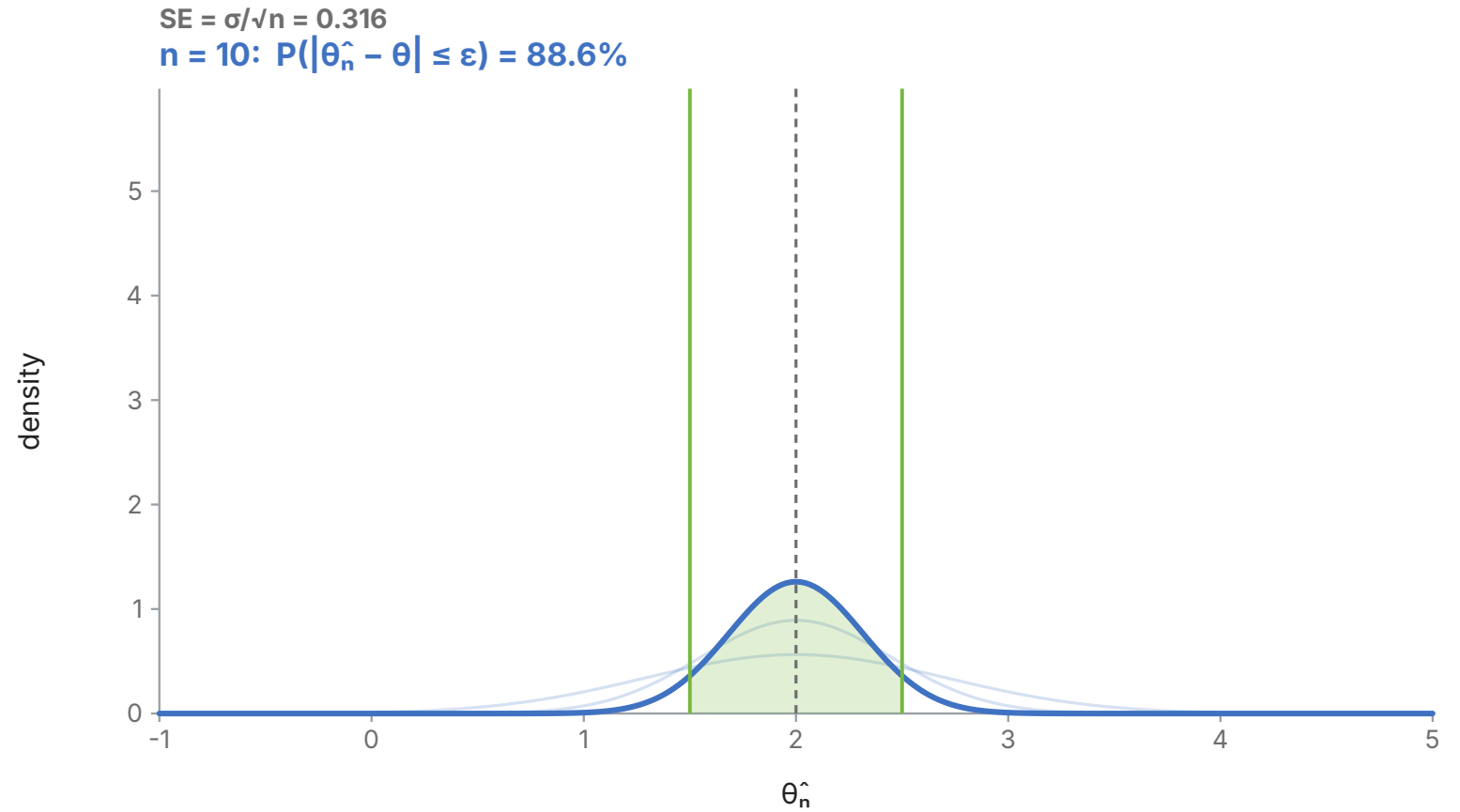
ϵ

0.5



The whole sampling distribution collapses onto θ ; the faint curves are the smaller sample sizes this one grew out of.

Consistency is the shaded percentage going to 100 for **every** ϵ , however small. It says nothing about how fast — that question returns with efficiency, later in the deck.



Estimation as Minimizing a Loss

What Does “Best” Mean?

Every estimator answers the question *which value fits the data best?* A **loss function** is how you say what “best” means.

Loss function

$L(y, c)$ is the cost of reporting c when the outcome is y . *How* it charges for a miss is your choice, not a fact about the data.

The recipe

Report the value with the smallest **average loss in your sample**:

$$\hat{\theta} = \arg \min_c \frac{1}{n} \sum_{i=1}^n L(y_i, c)$$

The sample average stands in for the population average $\mathbf{E}[L(Y, c)]$. The law of large numbers from the last section is why that substitution works as n grows.

Which Loss Leads Where

Loss $L(y, c)$	How a miss is charged	Best single value c	With regressors
Squared $(y - c)^2$	quadratically — big misses dominate	the mean	least squares (conditional mean)
Absolute $ y - c $	linearly — every unit the same	the median	median (LAD) regression
Asymmetric	τ per unit too low, $1 - \tau$ per unit too high	the τ -quantile	quantile regression
0-1 $1\{y \neq c\}$	every miss the same, however large	the mode	classification
Neg. log-likelihood $-\ln f(y; \theta)$	by how <i>surprising</i> y is under a model	the MLE	maximum likelihood

❗ **Choose the loss, and you have chosen the estimator**

The mean and the median answer *different* questions: the average income, or the typical one? The last row is the bridge to maximum likelihood.

Minimizing a Loss, Live

Loss

Squared ▾

candidate c

50



Add a top earner ☐

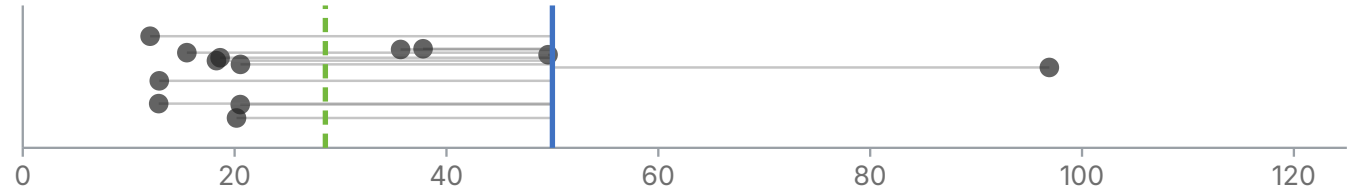
New sample

Top: 13 incomes and your candidate c ; the grey lines are the misses. Bottom: the average loss of reporting c , for every c . The green line is where it bottoms out.

Add the top earner: under squared loss the minimum jumps, under absolute loss it barely moves. Squared loss gives a far-away point a say proportional to its **squared** distance.

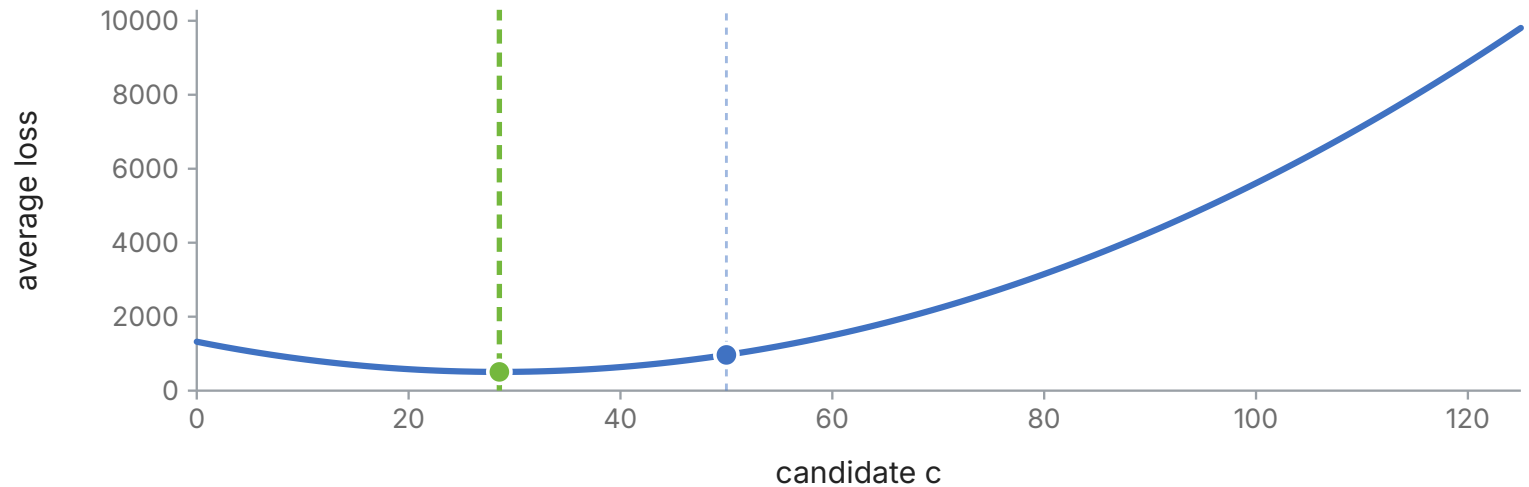
the sample: 13 incomes, in € 1000s

mean 28.6 median 20.2 90th percentile 49.6



minimised at $c = 28.6$: the sample mean

average loss at $c = 50.0$: 966 at the minimum: 507



What the Choice of Loss Buys

- **Squared loss** lets one large miss outweigh many small ones, so the mean chases outliers. It is the loss inside the **MSE** of the first section, and the one behind **least squares**
- **Absolute loss** charges every unit of a miss the same, so the median is **robust**: one top earner cannot drag it
- **Asymmetric loss** is for misses whose two directions cost differently: running out of stock vs. an unsold unit, a central bank under- vs. over-forecasting inflation

i The same recipe with regressors

Replace the constant c by a prediction $f(x_i; \theta)$ and minimize the same average loss over θ . Squared loss then fits the **conditional mean** $E[Y \mid X]$ — that is OLS. Absolute loss fits the **conditional median**.

Maximum Likelihood

The Idea: Make the Data Least Surprising

A loss needs no story about where the data came from. **Maximum likelihood** starts from one.

- Write down a **model**: a distribution $f(y; \theta)$ that says how probable each outcome is, for each value of θ
- For every candidate θ , ask: *how probable would the data I actually saw have been?*
- Report the θ under which that probability is **largest**

Maximum likelihood estimator

$$\hat{\theta}_{ML} = \arg \max_{\theta} \prod_{i=1}^n f(y_i; \theta) = \arg \max_{\theta} \sum_{i=1}^n \ln f(y_i; \theta)$$

Taking logs turns a product of tiny numbers into a sum; the maximizer does not move.

Likelihood, Live

We model n workers' log hourly wages as $N(\mu, \sigma^2)$. Which μ and σ make the wages we actually observed most probable?

candidate μ

2.6

candidate σ

0.8

observations n

8

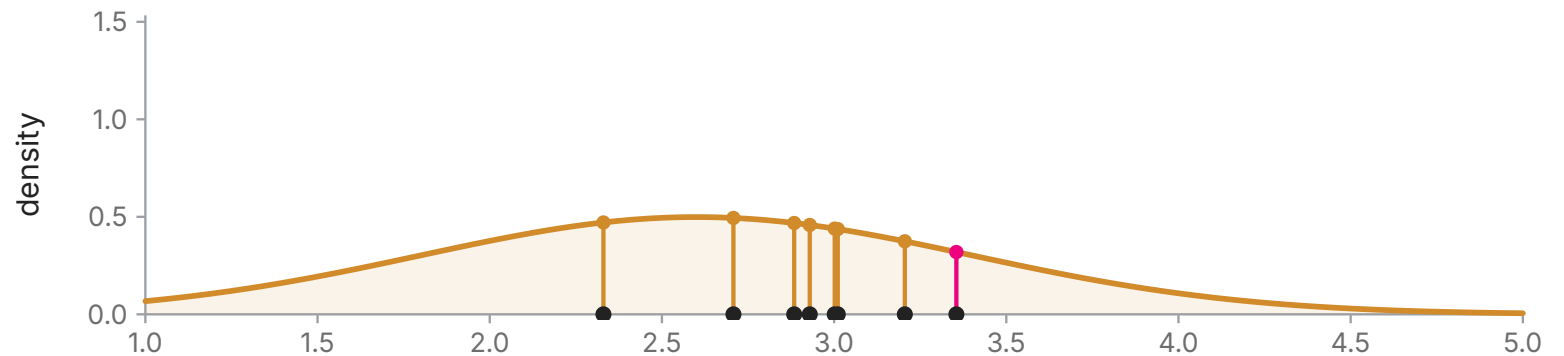
New sample

Each observation contributes the **height of the density** at its value — the stems. The likelihood multiplies them, so one point the curve barely reaches (pink) drags the whole product down.

Move μ and σ until the top picture matches the bottom one: that is maximum likelihood. Too narrow a density misses the outer points; too wide a one spreads itself thin over all of them.

your candidate: $\mu = 2.60$, $\sigma = 0.80$

likelihood = product of the 8 heights = 1.1×10^{-3} log-likelihood -6.8



best fit (MLE): $\mu = \bar{y} = 2.93$, $\sigma = 0.29$

likelihood = 0.226 log-likelihood -1.5 — no other μ , σ does better



Familiar Estimators Are MLEs

Data	Model	MLE	Remark
yes / no answers	Bernoulli(p)	$\hat{p} = k/n$	the share of yes answers
counts (patents, doctor visits)	Poisson(λ)	$\hat{\lambda} = \bar{y}$	the sample mean
durations (unemployment spells)	Exponential(λ)	$\hat{\lambda} = 1/\bar{y}$	a rate: one over the mean
continuous outcomes (log wages)	Normal(μ, σ^2)	$\hat{\mu} = \bar{y},$ $\hat{\sigma}^2 = \frac{1}{n} \sum (y_i - \bar{y})^2$	$\hat{\sigma}^2$ is the biased " $\div n$ " estimator from the first section — yet consistent

i Not exotic — but no longer a guess

For simple models, MLE returns the formulas you would have guessed, together with a *reason* for them. For models where there is nothing obvious to guess — logit and probit, duration and count regressions — it is the default recipe.

What Maximum Likelihood Assumes



What you must assume

- **The right model:** the data come from $f(y; \theta)$ — the whole distribution, not just its mean
- **Independence**, so the likelihood is a product
- **Identification:** different θ , different distributions
- **Regularity:** a smooth likelihood; the possible values of y do not depend on θ



In return, as n grows

- **Consistent:** $\hat{\theta} \rightarrow \theta$
- **Approximately normal** around θ
- **Efficient:** the smallest possible variance — made precise later
- **Invariant:** the MLE of $g(\theta)$ is $g(\hat{\theta})$



And if the model is wrong?

The MLE still converges — to the **best-fitting wrong model**. Sometimes harmless (the normal MLE of a mean is $\bar{y}$, whatever the true distribution); sometimes exactly the number you did not want.

Building the Bridge

Two Roads to the Same Estimator

Big picture: a loss and a likelihood both answer the same question — *which parameter values fit the data best?* They just define “best” differently.

- **Least squares (LS):** pick θ that minimizes the total squared prediction error — a *loss*, no model of the errors needed
- **Maximum likelihood (MLE):** pick θ that makes the observed data most probable — needs a full *model* of the errors

Key Insight

Under the right assumptions, these two different ideas lead to the **exact same estimator**. The next slide shows why.

From Maximum Likelihood to Least Squares

Suppose $Y = f(X; \theta) + \epsilon$ with $\epsilon \sim N(0, \sigma^2)$.

The likelihood of the data is a product of normal densities:

$$\text{MLE} = \arg \max_{\theta} \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - f(x_i; \theta))^2}{2\sigma^2}}$$

Taking logs and dropping the parts that don't involve θ ...

$$\text{MLE} = \arg \min_{\theta} \sum_{i=1}^n (y_i - f(x_i; \theta))^2 = \textbf{Least Squares}$$

i Takeaway

If the errors are normal, least squares *is* maximum likelihood — and inherits every guarantee on the MLE's list, efficiency included.

Every Likelihood Is a Loss

Maximizing $\sum \ln f$ is minimizing $\sum -\ln f$. So every error distribution you assume hands you a loss:

If you assume the error is...	then $-\ln f$ is, up to constants...	a loss called	and the MLE is
Normal	$(y - f(x; \theta))^2 / 2\sigma^2$	squared	least squares
Laplace (fatter tails)	$ y - f(x; \theta) / b$	absolute	median (LAD) regression
Bernoulli , $P(y = 1) = p(x; \theta)$	$-y \ln p - (1 - y) \ln(1 - p)$	log loss	logit / probit



Read it left to right

Assuming a distribution **chooses** your loss: MLE is loss minimization with a loss derived from a model.



Read it right to left

Choosing a loss **quietly assumes** a distribution. Squared loss suits roughly normal errors — and pays for it when tails are fat.

Generalized Method of Moments

From Method of Moments to GMM

A third road. Instead of a loss or a likelihood, write down facts that should hold *on average* at the true parameter, like $\mathbf{E}[Y] = \mu$. **Classical method of moments:** one such equation per parameter, solved exactly.

But what if you have *more* valid equations than parameters? You can't satisfy all of them exactly at once, so instead you get as close as possible to satisfying *all* of them simultaneously.

This is the idea behind GMM

- **Classical MM:** p parameters, p moment conditions (exactly identified)
- **GMM:** p parameters, $q \geq p$ moment conditions (over-identified)

The GMM Estimator

Suppose economic theory implies a set of conditions that should hold *on average* at the true parameter value:

$$\mathbf{E}[g(X_i, \theta)] = 0, \quad g : \mathbb{R}^d \times \mathbb{R}^p \rightarrow \mathbb{R}^q$$

Their sample counterparts will rarely hit zero exactly, so we make them **as close to zero as possible**:

$$\bar{g}_n(\theta) = \frac{1}{n} \sum_{i=1}^n g(X_i, \theta)$$

GMM Estimator

$$\hat{\theta}_{GMM} = \arg \min_{\theta} \bar{g}_n(\theta)' W_n \bar{g}_n(\theta)$$

where W_n is a $q \times q$ weight matrix that decides how much each condition counts.

How GMM Relates to the Others

Minimize a loss or maximize a likelihood, and the optimum sits where the **derivative is zero**. Averaged over the sample, that first-order condition *is* a moment condition.

Estimator	Built from	Its moment condition
Mean	squared loss	$E[Y - \mu] = 0$
Median	absolute loss	$P(Y \leq m) - \frac{1}{2} = 0$
OLS	squared loss, with regressors	$E[X(Y - X'\beta)] = 0$
MLE	a full distribution	$E\left[\frac{\partial}{\partial \theta} \ln f(Y; \theta)\right] = 0$
IV	nothing but the moment itself	$E[Z(Y - X'\beta)] = 0$

So GMM is the umbrella

Every estimator in this deck is GMM with a particular choice of moments. What differs is **how much you have to believe**: MLE needs the whole distribution, GMM only the moments you write down. The last row has no loss or likelihood behind it at all — that is where GMM earns its keep.

Choosing the Weights

Not all moment conditions are equally reliable — noisier ones should count for less.

i Two-Step Procedure

- (1) Start with equal weights ($W = I$) to get a first-pass estimate $\tilde{\theta}$
- (2) Use $\tilde{\theta}$ to see how noisy and correlated the moment conditions are
- (3) Re-estimate using weights that downweight the noisier conditions

With the optimal weight matrix, GMM achieves the **smallest possible variance** among all choices of W — an *efficiency* result, the theme of the next section.

For the curious: that smallest variance takes the compact form $V = (G'\Sigma^{-1}G)^{-1}$ — you'll see it derived in more advanced courses.

Example: Instrumental Variables Regression

Model: $y = X\beta + \epsilon$, but $E[X'\epsilon] \neq 0$ — X is **endogenous**, so OLS is biased.

Fix: find instruments Z that are correlated with X but uncorrelated with ϵ :

$$E[Z'\epsilon] = 0 \quad \implies \quad g(y, X, Z; \beta) = Z'(y - X\beta)$$

GMM / IV Estimator

$$\hat{\beta}_{GMM} = (X'ZW_nZ'X)^{-1}X'ZW_nZ'y$$

As many instruments as regressors: this is the classic IV estimator. More instruments than regressors: it's 2SLS with optimal weights.

OLS Bias vs. IV Consistency, Visualized

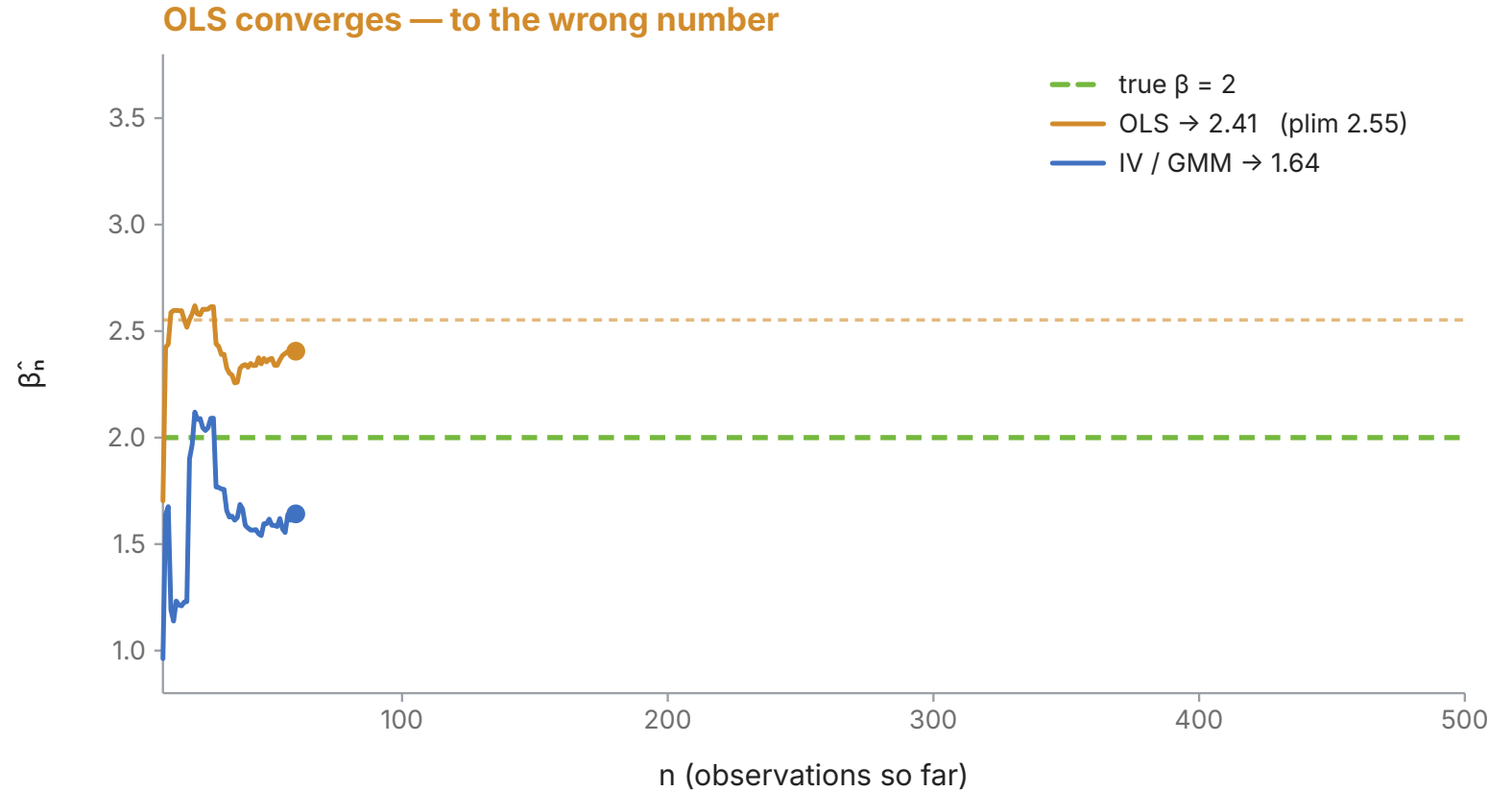
sample size n

60



A live simulation with $x = 0.9z + u$ and $y = 2x + u + \varepsilon$: the confounder u sits in both, so x is endogenous.

Both lines are noisy at small n and both settle down — but they settle on different numbers. More data does not cure a biased estimator; it only pins down the wrong answer more precisely.



GMM Shows Up All Over Economics

- **Instrumental variables & 2SLS:** fixing endogeneity in regression (the last two slides)
- **Rational expectations models:** Euler equations linking today's consumption choices to future returns
- **Asset pricing (CAPM):** relating an asset's expected return to its exposure to market risk
- In each case, economic theory supplies the moment conditions — GMM turns them into an estimator, without requiring a full distributional assumption

Model Testing and GMM's Family Tree

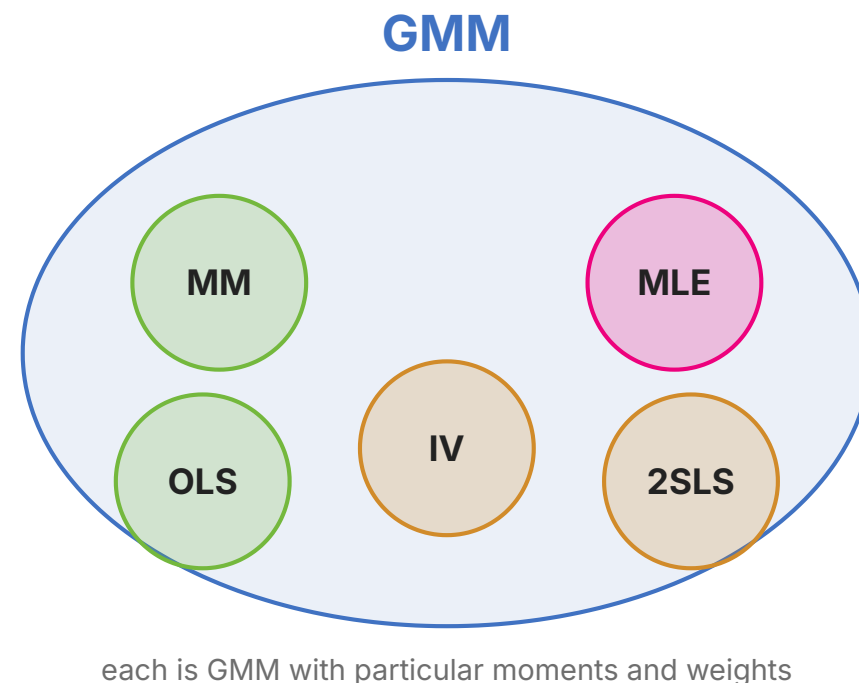
! J-Test for Over-Identification

When $q > p$, the extra moment conditions can be used to test the model itself:

$$J = n \cdot \bar{g}_n(\hat{\theta})' W \bar{g}_n(\hat{\theta}) \xrightarrow{d} \chi^2_{q-p}$$

A large J suggests some moment condition — and so the model — may be wrong.

Tests get their own lecture next. The end of this deck shows why every test is also an interval.



When (Not) to Reach for GMM



Good fit when...

- You have more moment conditions than parameters
- There's an endogeneity problem
- The model is nonlinear
- You don't want to assume a full distribution



Watch out for...

- GMM needs fairly large samples to behave well
- Results can be sensitive to *which* moments you pick
- Less efficient than a correct MLE: the price of assuming less

How Precise Is the Estimate?

That Variance Has a Name

Every sampling distribution in this deck had a width. For the sample mean it is $\text{Var}(\bar{X}) = \sigma^2/n$, and its square root is the number you report.

Standard deviation

$$\text{SD}(X_i) = \sqrt{\text{Var}(X_i)}$$

A property of the **population**. It does not shrink with n ; a larger sample only estimates it better.

Standard error

$$\text{SE}(T) = \sqrt{\widehat{\text{Var}}(T)}$$

A property of the **estimator** $T = T(X_1, \dots, X_n)$ — the width of the sampling distribution we have been drawing all along.

Every interval from here is built from the second one

$\bar{x} \pm 1.96 s$ says where the *data* lie; $\bar{x} \pm 1.96 s/\sqrt{n}$ says where the *mean* lies. The rest of this deck is about the second: how small it can get, and how to report it.

What Drives Estimation Uncertainty

For the sample mean, $SE = \sigma / \sqrt{n}$. The same three levers set the standard error of every estimator in this deck:

- **Noise in the data, σ :** noisier outcomes, a wider sampling distribution
- **Amount of data, $\sqrt{n}$:** the SE shrinks with the *square root* of n — four times the data to halve it
- **The estimator itself:** two estimators on the same data can have different SEs. How much of the data's information an estimator puts to use is its **efficiency**

i And the shape is (almost) always the same

In large samples the mean, least squares, MLE and GMM are all approximately **normal** around θ — under their assumptions — with that SE as their spread. Every interval at the end of this deck is built on this one fact.

Efficiency: How Low Can the Variance Go?

Among all unbiased estimators, some are more **precise** than others. **Efficiency** asks: is $\hat{\theta}$ the most precise **unbiased** estimator possible given some sample size?

It turns out there is a hard floor on how small the variance of an unbiased estimator can ever be — no amount of cleverness can beat it.

That floor is the Cramér-Rao Lower Bound.

The Cramér–Rao Lower Bound

Fisher Information

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2 \ln f(X; \theta)}{\partial \theta^2} \right]$$

Roughly: how sharply peaked the likelihood is — how fast the log-likelihood in *Likelihood, Live* falls as you move away from the best fit. More information means the data pin down θ more precisely.

Cramér–Rao Lower Bound (CRLB)

For *any* unbiased estimator $\hat{\theta}$ based on n observations:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

An unbiased estimator that *achieves* this bound is called **efficient**. The MLE reaches it as n grows.

Example: The Sample Mean Is Efficient

For $X_i \sim N(\mu, \sigma^2)$, estimating μ :

- Fisher information: $I(\mu) = 1/\sigma^2$
- Cramér-Rao bound: $\text{CRLB} = \sigma^2/n$
- Actual variance: $\text{Var}(\bar{X}) = \sigma^2/n$

Conclusion

$\text{Var}(\bar{X})$ exactly meets the CRLB, so $\bar{X}$ is an **efficient** estimator of μ — no unbiased estimator can do better.

Efficiency, Visualized

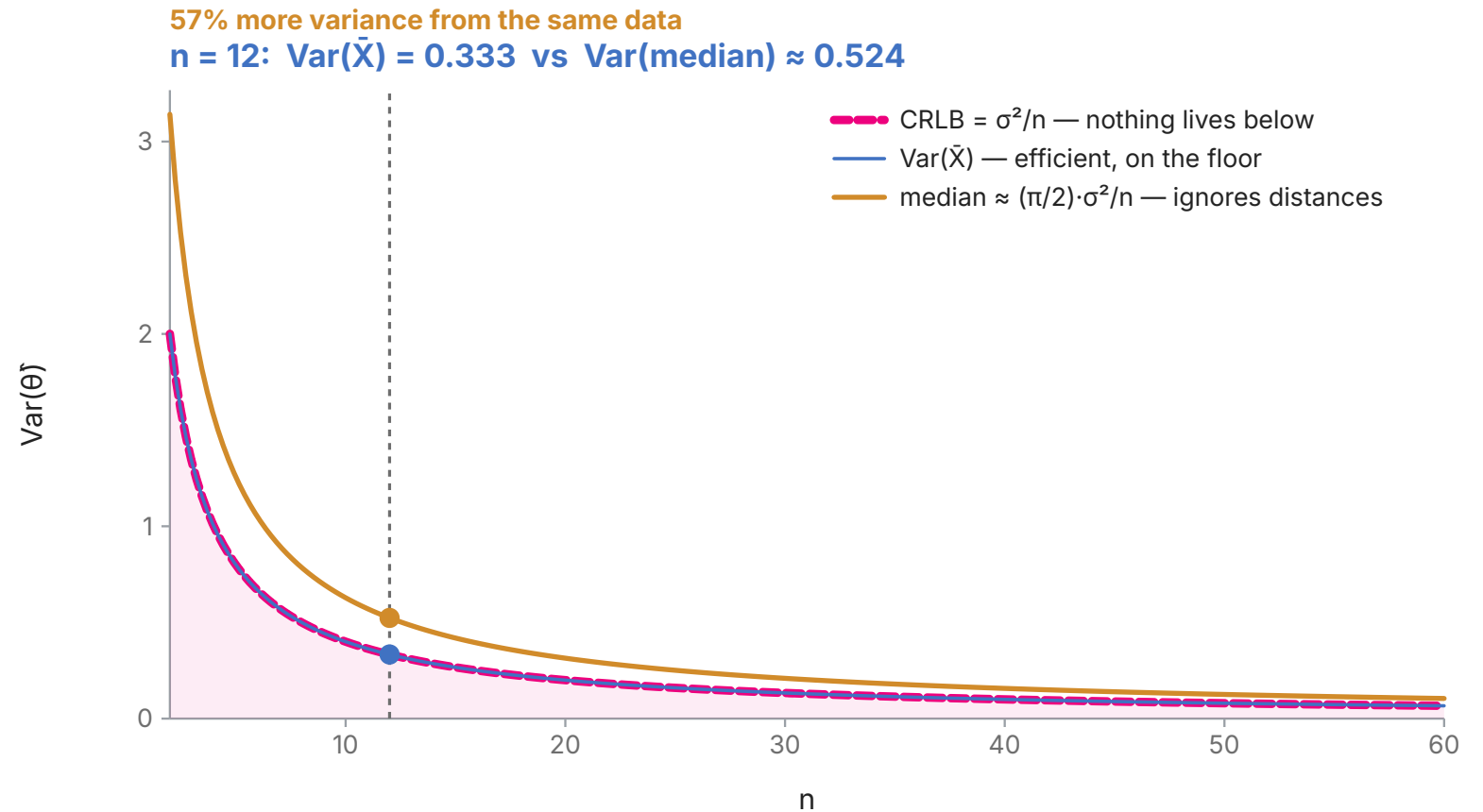
n

12



$\text{Var}(\bar{X})$ sits exactly on the CRLB floor; the shaded region below it is empty for **every** unbiased estimator, at every n .

The **median** is unbiased here too — the normal is symmetric — but it ignores *how far* each point lies from the centre. In large samples its variance is $\pi/2 \approx 1.57$ times the floor: it needs 57% more data for the same precision.



Mean or Median? Efficiency, Live

share of wild points

0



sample size n

25



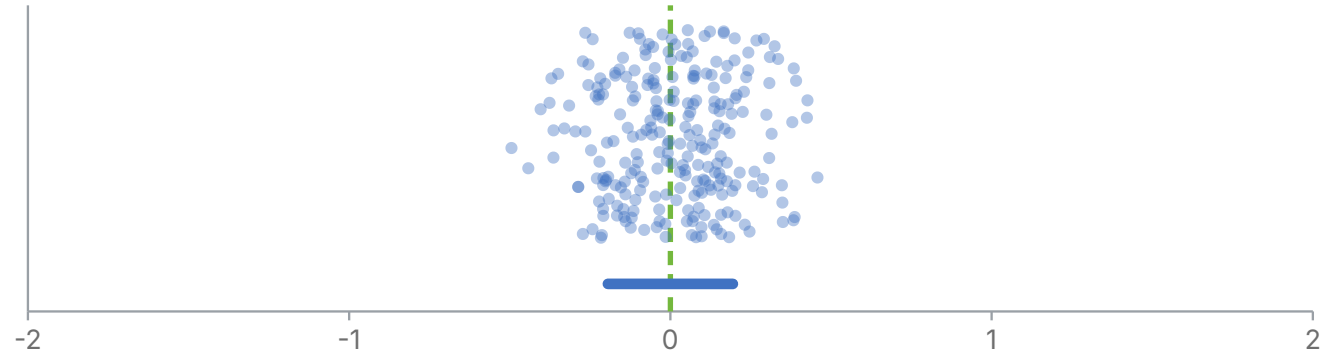
New simulations

Each dot is one estimate from a fresh sample; the truth is 0. Most points come from $N(0, 1)$, a share of “wild” ones from a distribution ten times as wide. Both are centred on 0, so both estimators stay unbiased.

On clean data the mean wins, by roughly the $\pi/2$ of the last slide. A few percent of wild points flips the ranking: the mean inherits their variance, the median barely notices. Efficiency is a property of an estimator **and** the data it meets.

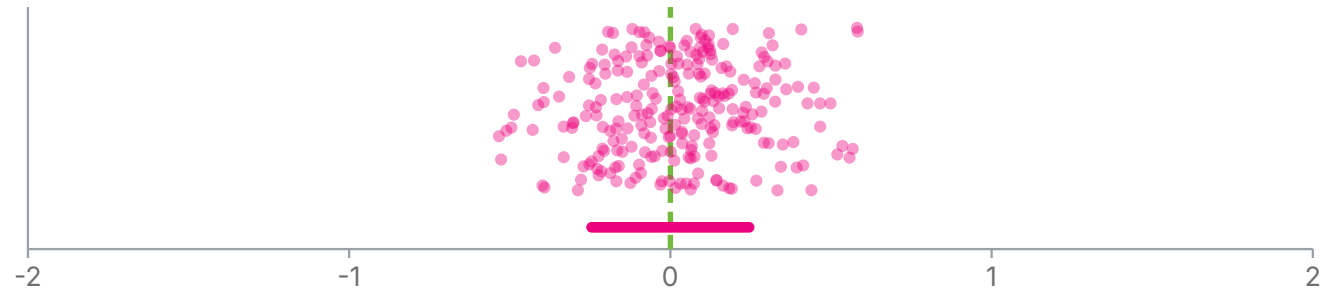
clean $N(0, 1)$ data, $n = 25$
SD 0.195 (bar: ± 1 SD)

mean



$\text{Var}(\text{median}) \div \text{Var}(\text{mean}) = 1.59 \rightarrow$ the mean is more efficient
SD 0.245 (bar: ± 1 SD)

median



estimate (truth = 0)

Efficiency Is Bought with Assumptions

- **MLE** reaches the Cramér-Rao floor as n grows — *if* the distribution you assumed is the true one
- **Least squares** is the best *linear* unbiased estimator (Gauss-Markov), and fully efficient when errors are normal, because then it *is* the MLE
- **GMM** makes the best use of the moments you gave it, and ignores everything else — so a correct MLE beats it
- **The median** loses to the mean on normal data and beats it when tails are fat



The trade-off behind every choice in this deck

The more you assume, the more precise you can be — **if** you are right. Assume less, and you trade precision for robustness.



What is the variance *for*?

A variance nobody reports is useless. Turning the standard error into a statement on a page is the job of the **confidence interval**.

From Spread to Interval

From a Point to a Range

A point estimate says *what is our best guess*, never *how good the guess is* — the question a policy note has to survive.

Confidence interval

An **interval estimator** $[\hat{\theta}_L, \hat{\theta}_U]$ has **confidence level** $1 - \alpha$ if its **coverage probability** $P_{\theta}(\hat{\theta}_L \leq \theta \leq \hat{\theta}_U)$ is $1 - \alpha$, whatever the true θ .

The workhorse

The estimator is approximately normal, so $\hat{\theta} \pm 1.96 \times \text{SE}(\hat{\theta})$ covers θ in about 95% of samples. A small-sample mean with σ estimated by s uses t_{n-1} instead of 1.96.

Read the probability carefully

$\hat{\theta}_L$ and $\hat{\theta}_U$ are the random variables. θ is a fixed unknown constant. The probability is over **intervals**, not over θ .

Coverage, Live

Each horizontal line is one 95% interval from one sample — the same repeated sampling as the dot strips earlier, with a bar instead of a dot. The vertical rule is the true μ , which in real life you never see.

Sample size n

20



Level

95%

Population

Normal

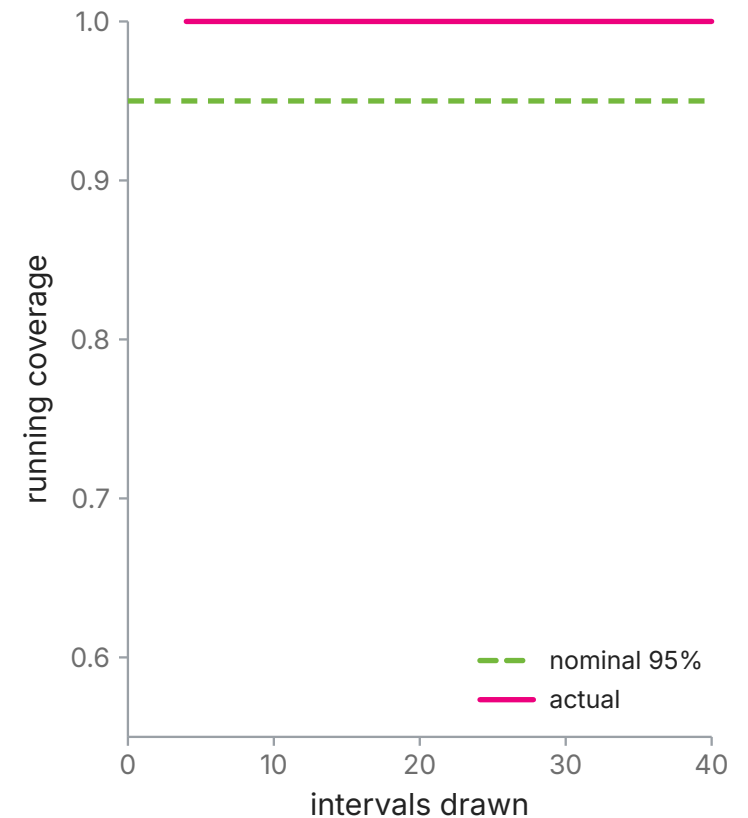
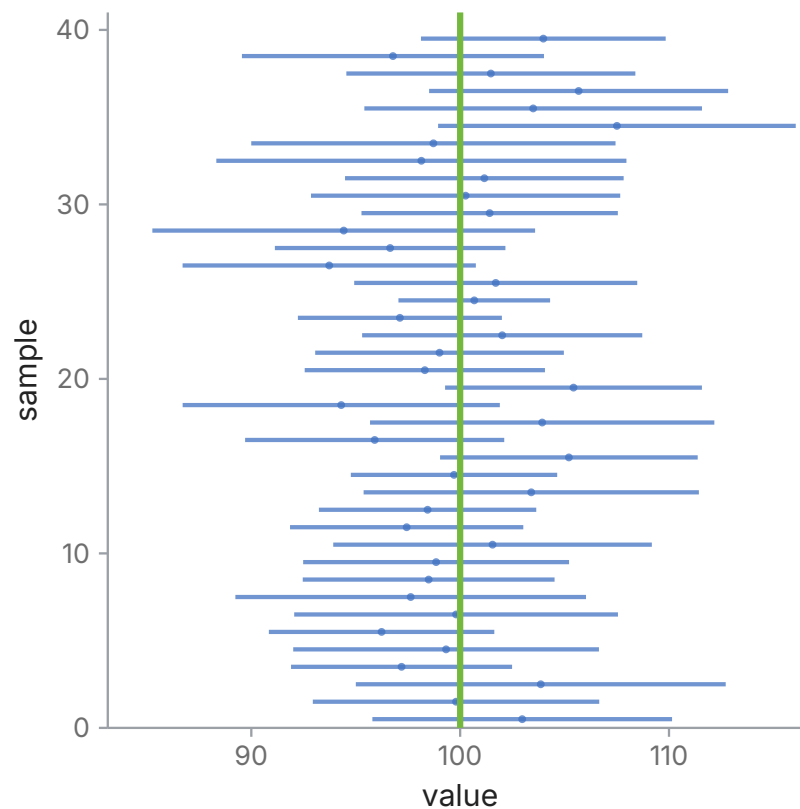
Method

t interval (correct)

Draw 40 more

95% is a promise about the **procedure**, not about the interval on your screen. Break an assumption — a skewed population at small n , a rare binary event, or z where t belongs — and the promise quietly stops being kept.

nominal 95% → actual 100.0% (40 of 40)
last 40 intervals — 0 miss



What the Interval Plot Shows

- With the t interval on a normal population, the running coverage settles on the nominal level — the procedure keeps its promise
- A **skewed** population at $n = 10$ under-covers: the interval is symmetric, the sampling distribution is not
- **Bernoulli with $p = 0.05$** is worse still, and the intervals visibly extend below zero
- Substituting z for t at small n under-covers — which is precisely why t exists



The point

Nominal coverage is a property of the **procedure under its assumptions**. The label keeps saying 95% also if the assumptions are violated.

The scatter of the dots in *Unbiasedness, Visualized* and the length of these bars are the same quantity: one standard error, times a critical value.

What a Confidence Interval Is *Not*



Correct interpretation

- 95% of intervals **built this way** contain θ
- The procedure fails 5% of the time, in the long run
- Wider interval $\Rightarrow$ less precise estimate



A confidence interval is NOT

- $P(\theta \in [\hat{\theta}_L, \hat{\theta}_U] \mid \text{data}) = 0.95$
- A range containing 95% of the **data**
- A range containing 95% of **future estimates**
- "They overlap, so the difference is not significant"

Once you have seen the data, the interval either contains θ or it does not. Nothing random is left to attach a probability to.

The third item is a genuinely different number: about 83% of future estimates fall inside a given 95% interval.

Confidence Interval $\neq$ Prediction Interval

Two intervals that look identical on a chart and answer different questions.



Confidence interval — for the mean

$$\bar{x} \pm z \frac{\sigma}{\sqrt{n}} \xrightarrow{n \rightarrow \infty} \{\mu\}$$

Width $\rightarrow 0$. With enough data you know μ exactly.



Prediction interval — for the next draw

$$\bar{x} \pm z \sigma \sqrt{1 + \frac{1}{n}} \xrightarrow{n \rightarrow \infty} \mu \pm z \sigma$$

Width $\rightarrow 2z\sigma$. It **stops**.



The distinction economists get wrong

Forecast fan charts are prediction intervals. Reading one as a confidence interval understates forecast uncertainty by a factor that grows with $\sqrt{n}$. The left-hand limit *is* the **consistency** picture from earlier; the right-hand one cannot collapse, because the next observation is not an estimator of anything.

p-Values and Intervals

What a p-Value Measures

Turn the question around. Instead of *which values of θ are plausible?*, ask about **one** value θ_0 — say “the reform had no effect”, $\theta_0 = 0$.

p-value, in words

If θ really were θ_0 , how often would a fresh sample give an estimate **at least as far** from θ_0 as the one we got?

$$p(\theta_0) = P_{\theta_0}(|\hat{\theta} - \theta_0| \geq |\hat{\theta}_{\text{obs}} - \theta_0|)$$

The sampling distribution again — this time centred at θ_0 .



Small p

The data would be **surprising** if θ were θ_0 : evidence against it.
Below a threshold α , often 5%, we *reject* θ_0 .



Not

The probability that θ_0 is true — θ_0 is either true or not. The next lecture treats tests and p-values in full.

A Confidence Interval Is an Inverted Test

Hypothesis testing, the next lecture, builds rejection regions; this deck builds intervals. They are the same object, seen from two sides.

Duality

$$\text{CI}_{1-\alpha}(x) = \{ \theta_0 : \text{the level-}\alpha \text{ test of } H_0: \theta = \theta_0 \text{ does not reject} \}$$

Equivalently, writing $p(\theta_0)$ for the p-value of the test of $H_0 : \theta = \theta_0$,

$$\text{CI}_{1-\alpha}(x) = \{ \theta_0 : p(\theta_0) > \alpha \}$$

Read that as a picture

The confidence interval is the **horizontal slice** of the p-value curve at height α .

The p-Value Curve, Live

Drag θ_0 and watch the test on the left agree with the interval on the right — every time.

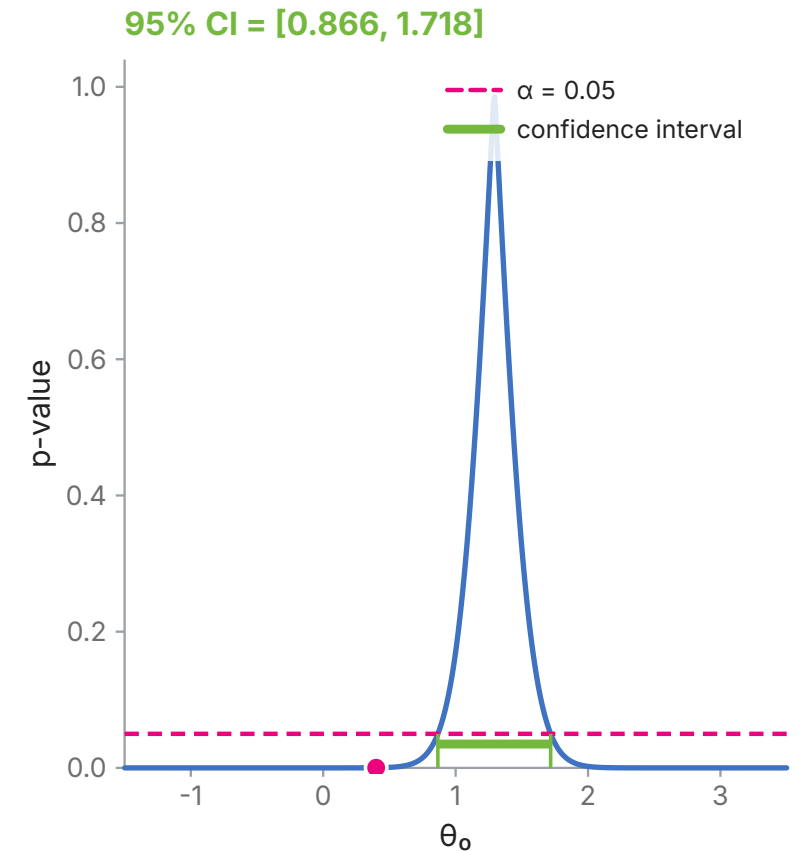
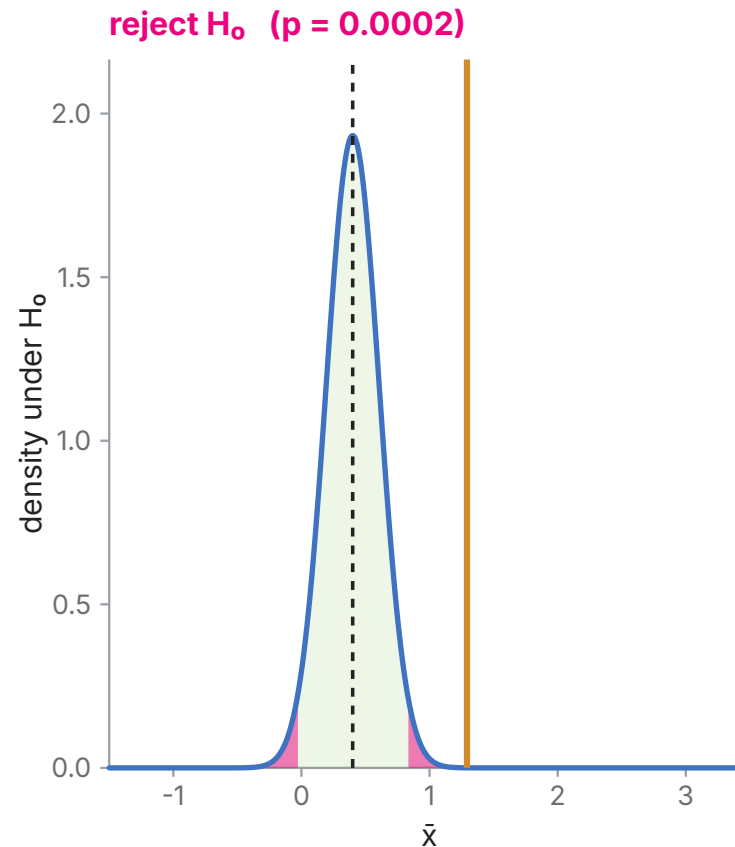
θ_0
 

α
 

n
 

Left: the null distribution centred at the current θ_0 , with the observed $\bar{x}$ and its tail area.

Right: $p(\theta_0)$ for **every** θ_0 . Where the curve crosses α are exactly the interval endpoints.



What the Duality Buys You

- You never need two sets of tables. "Is θ_0 in the interval?" and "is $p < \alpha$?" are one question
- Lower α and the slice moves down, so the interval **widens** — the same trade-off a rejection region draws
- The interval says strictly more than the test: it also reports every value the data **cannot** rule out

! Why this matters for reporting

"Not significant" collapses an interval to one bit. An interval of $[-0.01, 0.42]$ and an interval of $[-0.20, 0.21]$ tell very different stories, and the p-value is nearly the same for both.

When the Formula Runs Out

The Luxury You Only Have in a Lecture

Every simulation so far drew fresh samples **from a known truth** — a luxury you have exactly once, in a lecture.

! You cannot do that with real data

One sample, no population to redraw from. The way out so far was a **formula** for the standard error; these have none:

- The **Gini coefficient**, the 90–10 ratio, top income shares
- **Elasticities**, ratios of estimates, differences of **quantiles**
- Anything computed by a multi-step procedure

i The bootstrap principle

The sample is to the population as a **resample** is to the sample. Draw B samples of size n **with replacement** from the data, recompute $\hat{\theta}^*$ on each, and use the spread of $\hat{\theta}^*$ as the sampling distribution of $\hat{\theta}$.

The Bootstrap, Live

Statistic

Gini coefficient ▾

Sample size n

80

Replicates B

600

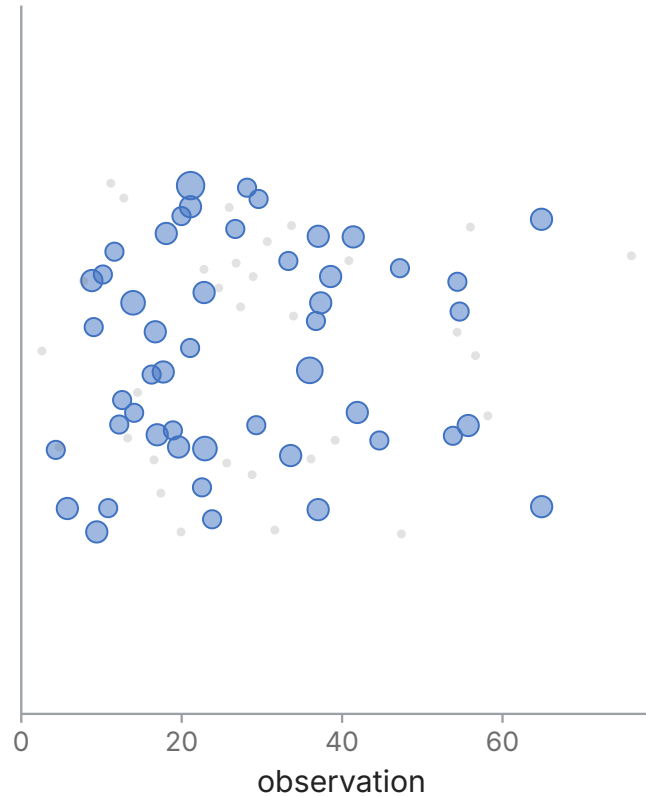
New sample

Left: the sample. Dot **size** is how often each observation was drawn in one replicate; greyed-out points were not drawn at all.

Choose **Maximum** to watch the method fail: a resample misses the largest observation with probability $(1 - 1/n)^n \rightarrow e^{-1}$.

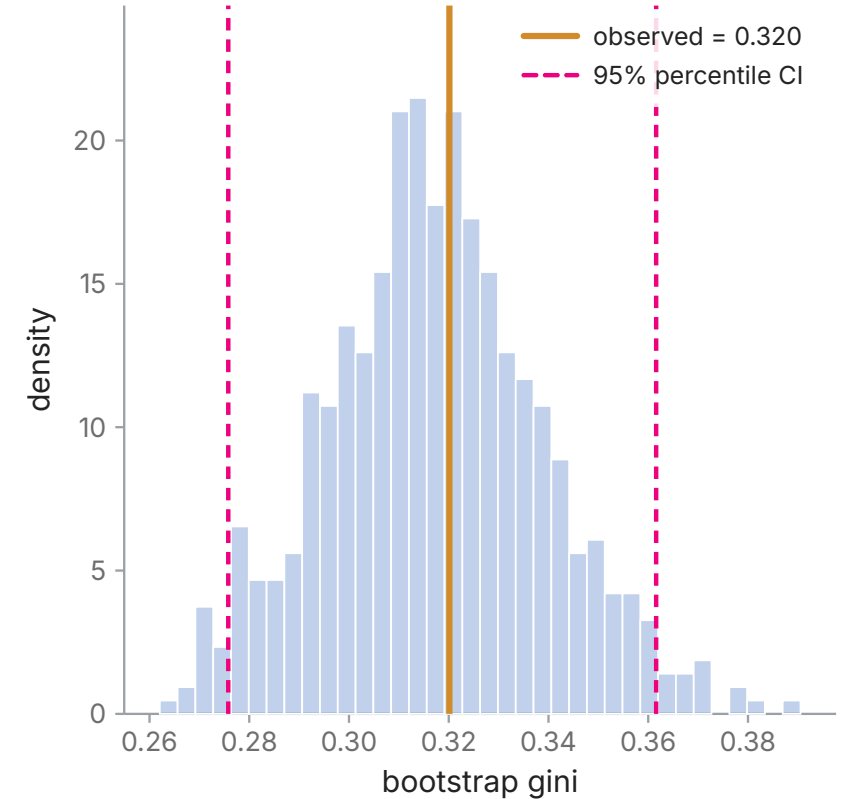
the sample

one replicate: 31 of 80 never drawn



95% CI [0.276, 0.362]

bootstrap SE 0.0216 — no textbook formula exists



What the Bootstrap Can and Cannot Do

- For the **mean**, the bootstrap standard error reproduces $s/\sqrt{n}$ to two decimals — which is what earns it trust
- For the **Gini**, it produces an interval where no textbook formula exists at all
- For the **maximum** it collapses: a resample misses the largest observation with probability $(1 - 1/n)^n \rightarrow e^{-1} \approx 0.368$, so roughly 63% of replicates return exactly the sample max

! *B* is not *n*

Raising *B* shrinks Monte-Carlo noise in the **endpoints**. It does not shrink the interval, and it adds no information.

! It cannot repair the sample

Resampling a biased sample gives biased resamples. And i.i.d. resampling destroys dependence — clustered or serial data need a block bootstrap.

Summary

Comparing the Estimation Methods

	Least squares	Median / LAD	MLE	GMM
Built from	squared loss	absolute loss	a full distribution	moment conditions
Needs a full distribution	✗	✗	✓	✗
Robust to outliers	✗	✓	depends on model	depends on moments
Efficient when...	errors normal	errors Laplace	the model is right	best weights, given its moments



Rules of Thumb

- **Quick and simple:** method of moments
- **The average outcome, no distribution needed:** least squares
- **The typical outcome, or outliers in the data:** median / LAD
- **You know the distribution:** MLE
- **Endogeneity or extra moment conditions:** GMM

Key Takeaways – The Estimator

- (1) An estimator is judged by its **sampling distribution**: where it sits (bias), how wide it is (variance), and whether it closes in on θ (consistency)
- (2) **MSE = variance + bias²** — which is why a little bias can be worth buying
- (3) **Choose a loss, get an estimator**: squared $\rightarrow$ mean, absolute $\rightarrow$ median, asymmetric $\rightarrow$ quantile
- (4) **MLE** is loss minimization with the loss derived from a model; with normal errors it *is* least squares
- (5) **GMM** needs only moment conditions — and every estimator above is a special case



The through-line

Every one of these is a statement about the **same** distribution — the one you never observe, because you only ever draw one sample from it.

Key Takeaways – Uncertainty

- (1) The **standard error** is the width of the sampling distribution; it shrinks like $1/\sqrt{n}$
- (2) **Efficiency** is how little of the data's information an estimator wastes — bought with assumptions, lost when they fail
- (3) A confidence interval is a statement about the **procedure**; coverage is kept only while its assumptions hold
- (4) A **p-value** asks how surprising the data are under one θ_0 ; the interval is every θ_0 that is not surprising
- (5) The **bootstrap** covers statistics with no formula — and fails visibly at the edge of the sample



Choosing an interval

- **Large sample** — $\hat{\theta} \pm 1.96 \text{ SE}$
- **Mean, small sample, unknown σ** — t interval
- **Smooth function of estimates** — delta method
- **No formula** — bootstrap



Report the interval

It carries the effect size, the precision, and every value the data cannot rule out. A p-value carries one bit of that.

What This Deck Left Out

Bayesian methods

Priors and posteriors · **credible** intervals, the probability statement a CI is not · hierarchical models

Finite-sample optimality

Sufficiency · Rao-Blackwell · the **UMVUE**, the best unbiased estimator

Robust and modern estimation

M-estimators, the general form of the loss section · LASSO and Ridge · machine learning and causal inference



And the interval kind

Pivots and exact intervals — χ^2 for a variance, Wilson for a proportion · sample-size planning · the **delta method** · cluster bootstraps